

STD 1: Measurement (Std 1), M1 Applications of Measurement (Y11)

Units and Measurement Error (Std 1)

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Exam Equivalent Time: 37.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-M1 Applications of Measurement* has been examined in a feast or famine manner in the first two *Std1 HSC exams*. In 2020, it attracted a solitary 1-mark multiple choice question as opposed to 2019 where it was allocated a full 10 marks (12.5% of the exam) across 3 separate questions.
- This analysis looks at the sub-topic *Units and Measurement Error*.

ANALYSIS - What to expect and common pitfalls

- *Units and Measurement Error* was allocated a single multiple choice question in the *2020 Std1 exam* after being omitted from the *2019 Std1 exam*.
- *Measurement Error*, although not examined in Std1 tests to date, remains a likely candidate going forward and deserves revision attention. Our questions and solutions have been adjusted to reflect the "simplified" new syllabus calculation of percentage error (as a positive number, not the previous $\pm$ expression).
- *Significant figures* and *scientific notation* were combined in *2020 Std1 3 MC* and produced a big red flag with a mean mark of just 12%!
- *Significant figures* has form in producing surprisingly low mean marks with *2015 Std2 1 MC* resulting in the lowest mean mark of all multiple choice that year (not a typo, the *first* question in the exam).
- *Scientific Notation* questions have produced volatile mean marks in the past. Pay careful attention to *2009 Std2 25b* which produced a state mean mark of 20%!
- Converting between units (eg. kilograms $\rightarrow$ grams) is often required. Few students were able to convert $\text{cm}^2 \rightarrow \text{m}^2$ which was required in *2009 Std2 12 MC* and is recommended revision.
- NESA Nov-19 syllabus maintenance includes the following in standard prefixes: mega-, giga-, nano- and micro-. These are examinable from the *2021 Std1 exam* going forward.

Questions

1. Measurement, STD2 M1 2007 HSC 1 MC

What is **0.000 000 326** mm expressed in scientific notation?

- (A) $0.326 \times 10^{-6} \text{ mm}$
- (B) $3.26 \times 10^{-7} \text{ mm}$
- (C) $0.326 \times 10^6 \text{ mm}$
- (D) $3.26 \times 10^7 \text{ mm}$

2. Measurement, STD2 M1 2014 HSC 2 MC

A measurement of 72 cm is increased by 20% and then the result is decreased by 20%.

What is the new measurement, correct to the nearest centimetre?

- (A) 46 cm
- (B) 69 cm
- (C) 72 cm
- (D) 104 cm

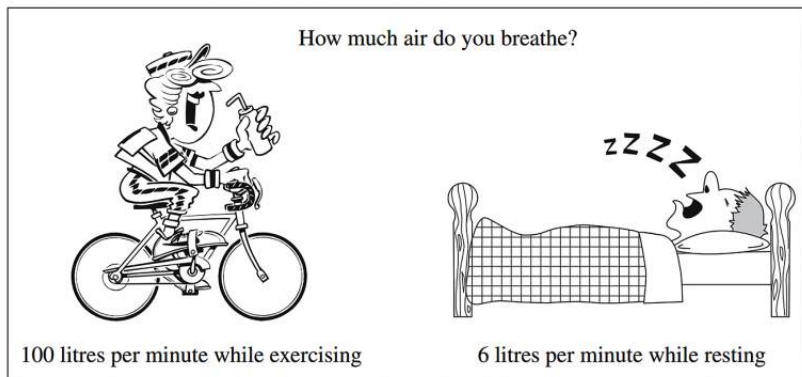
3. Measurement, STD2 M1 SM-Bank 25 MC

A cockroach is measured in a school science experiment and its length is recorded as 5.2 cm.

What is the upper limit of accuracy of this measurement?

- A. 5:21 cm
- B. 5.25 cm
- C. 5.5 cm
- D. 5.9 cm

4. Measurement, STD2 M1 2004 HSC 13 MC



During a ten-minute period, Kath is exercising and Jim is resting.

How much more air would Kath breathe than Jim during this time?

- (A) 40 Litres
- (B) 94 Litres
- (C) 940 Litres
- (D) 1060 Litres

5. Measurement, STD2 M1 2006 HSC 11 MC

Peter rides his bike at a speed of 27 km/h.

What is this speed in m/s?

- (A) 7.5
- (B) 18.75
- (C) 97.2
- (D) 450

6. Measurement, STD2 M1 2018 HSC 18 MC

The length of a window is measured as 2.4 m.

Which calculation will give the percentage error for this measurement?

- A. $\frac{0.05}{2.4} \times 100$
- B. $\frac{0.05}{100} \times 2.4$
- C. $\frac{0.5}{2.4} \times 100$
- D. $\frac{0.5}{100} \times 2.4$

7. Measurement, STD2 M1 2015 HSC 1 MC

What is 1 560 200 km written in standard form correct to two significant figures?

- (A) $1.56 \times 10^4 \text{ km}$
- (B) $1.6 \times 10^5 \text{ km}$
- (C) $1.56 \times 10^6 \text{ km}$
- (D) $1.6 \times 10^6 \text{ km}$

8. Measurement, STD2 M1 2016 HSC 1 MC

What is 208.345 correct to two significant figures?

- (A) 208
- (B) 210
- (C) 208.34
- (D) 208.35

9. Measurement, STD2 M1 2014 HSC 10 MC

The top of the Sydney Harbour Bridge is measured to be 138.4 m above sea level.

What is the percentage error in this measurement?

- (A) 0.036%
- (B) 0.050%
- (C) 0.072%
- (D) 0.289%

10. Measurement, STD2 M1 2015 HSC 12 MC

The length of a fish was measured to be 49 cm, correct to the nearest cm.

What is the percentage error in this measurement, correct to one significant figure?

- (A) 0.01%
 - (B) 0.5%
 - (C) 1%
 - (D) 2%
-

11. Measurement, STD2 M1 2017 HSC 21 MC

The length of a netball court is measured to be 30.50 metres, correct to the nearest centimetre.

What is the lower limit for the length of the netball court?

- A. 30.45 m
 - B. 30.49 m
 - C. 30.495 m
 - D. 30.499 m
-

12. Measurement, STD1 M1 2021 HSC 7 MC

Suppose $a = \frac{b}{7}$, where $b = 22$.

What is the value of a , correct to three significant figures?

- A. 3.14
 - B. 3.15
 - C. 3.142
 - D. 3.143
-

13. Measurement, STD2 M1 2019 HSC 8 MC

A person's weight is measured as 79.3 kg.

What is the absolute error of this measurement?

- A. 10 grams
 - B. 50 grams
 - C. 100 grams
 - D. 500 grams
-

14. Measurement, STD2 M1 2020 HSC 2 MC

What is 0.002073 expressed in standard form with two significant figures?

- A. 2.07×10^{-2}
 - B. 2.1×10^{-2}
 - C. 2.07×10^{-3}
 - D. 2.1×10^{-3}
-

15. Measurement, STD2 M1 2020 HSC 5 MC

A plant stem is measured to be 16.0 cm, correct to one decimal place.

What is the percentage error in this measurement?

- A. 0.3125%
 - B. 0.625%
 - C. 3.125%
 - D. 6.25%
-

16. Measurement, STD2 M1 2009 HSC 12 MC

How many square centimetres are in 0.0075 square metres?

- (A) 0.75
 - (B) 7.5
 - (C) 75
 - (D) 7500
-

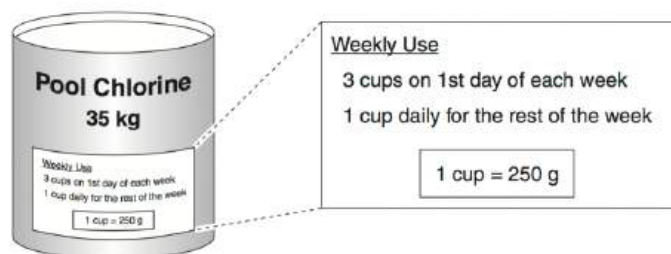
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18. Measurement, STD2 M1 2012 HSC 26g

Bhawana purchases pool chlorine in a new container which holds 35 kg.



She begins using this new container on the first day of a week.

How many full weeks should this container last? (2 marks)

19. Measurement, STD2 M1 2008 HSC 23b

The capacity of a bottle is measured as 1.25 litres correct to the nearest 10 millilitres.

What is the percentage error for this measurement? (1 mark)

20. Measurement, STD2 M1 2013 HSC 27d

A rectangular wooden chopping board is advertised as being 17 cm by 25 cm, with each side measured to the nearest centimetre.

- Calculate the percentage error in the measurement of the longer side. (1 mark)
- Between what lower and upper limits does the actual area of the top of the chopping board lie? (2 marks)

21. Measurement, STD2 M1 2009 HSC 25b

The mass of a sample of microbes is 50 mg. There are approximately 2.5×10^6 microbes in the sample.

In standard form, what is the approximate mass in grams of one microbe? (2 marks)

Worked Solutions

1. Measurement, STD2 M1 2007 HSC 1 MC

$$\begin{aligned} &0.000\,000\,326\text{ mm} \\ &= 3.26 \times 10^{-7}\text{ mm} \\ &\Rightarrow B \end{aligned}$$

2. Measurement, STD2 M1 2014 HSC 2 MC

$$\begin{aligned} &72 \text{ increased by } 20\% \\ &= 72 + (20\% \times 72) = 86.4\text{ cm} \\ &86.4 \text{ decreased by } 20\% \\ &= 86.4 - (20\% \times 86.4) = 69.12\text{ cm} \\ &\Rightarrow B \end{aligned}$$

STRATEGY: Students confident in this area could save time in the calculations as follows:
 $72 \times 1.2 \times 0.8 = 69.12$

3. Measurement, STD2 M1 SM-Bank 25 MC

$$\begin{aligned} \text{Absolute error} &= 0.05\text{ cm} \\ \text{Upper limit} &= 5.2 + 0.05 \\ &= 5.25\text{ cm} \\ &\Rightarrow B \end{aligned}$$

4. Measurement, STD2 M1 2004 HSC 13 MC

$$\begin{aligned} \text{Kath's air volume} &= 10 \times 100 \\ &= 1000\text{ L} \\ \text{Jim's air volume} &= 10 \times 6 \\ &= 60\text{ L} \\ \therefore \text{Extra air that Kath breathes} &= 1000 - 60 \\ &= 940\text{ L} \\ &\Rightarrow C \end{aligned}$$

5. Measurement, STD2 M1 2006 HSC 11 MC

$$\begin{aligned}
 27 \text{ km/h} &= 27\,000 \text{ metres per hour} \\
 &= \frac{27\,000}{60} \text{ metres per minute} \\
 &= \frac{27\,000}{60 \times 60} \text{ metres per second} \\
 &= 7.5 \text{ m/s} \\
 \Rightarrow A
 \end{aligned}$$

6. Measurement, STD2 M1 2018 HSC 18 MC

$$\begin{aligned}
 \text{Absolute error} &= 0.05 \text{ m} \\
 \% \text{ error} &= \frac{0.05}{2.4} \times 100 \\
 \Rightarrow A
 \end{aligned}$$

7. Measurement, STD2 M1 2015 HSC 1 MC

$$\begin{aligned}
 1\,560\,200 \\
 &= 1.5602 \times 10^6 \\
 &= 1.6 \times 10^6 \text{ km (2 sig)} \\
 \Rightarrow D
 \end{aligned}$$

♦♦ Mean mark 30%.
COMMENT: Incredibly, the first MC question in 2015 had the lowest mean mark of all MC questions in the exam!

8. Measurement, STD2 M1 2016 HSC 1 MC

$$\begin{aligned}
 208.345 &= 210 \text{ (2 sig.)} \\
 \Rightarrow B
 \end{aligned}$$

♦♦ Mean mark 36%!!

9. Measurement, STD2 M1 2014 HSC 10 MC

$$\begin{aligned}
 \text{Absolute error} &= 0.05 \text{ m} \\
 \% \text{ error} &= \frac{0.05}{138.4} \times 100 \\
 &= 0.036\% \\
 \Rightarrow A
 \end{aligned}$$

♦ Mean mark 48%

10. Measurement, STD2 M1 2015 HSC 12 MC

$$\begin{aligned}
 \text{Absolute error} &= 0.5 \text{ cm} \\
 \therefore \% \text{ error} &= \frac{0.5}{49} \\
 &= 1\% \\
 \Rightarrow C
 \end{aligned}$$

♦ Mean mark 41%.

11. Measurement, STD2 M1 2017 HSC 21 MC

$$\begin{aligned}
 \text{Absolute error} &= 0.5 \text{ cm} \\
 &= 0.005 \text{ m} \\
 \therefore \text{Lower limit} &= 30.50 - 0.005 \\
 &= 30.495 \text{ m} \\
 \Rightarrow C
 \end{aligned}$$

12. Measurement, STD1 M1 2021 HSC 7 MC

$$\begin{aligned}
 a &= \frac{b}{7} = \frac{22}{7} = 3.1428... \\
 3.1428 &= 3.14 \text{ (to 3 sig fig)} \\
 \Rightarrow A
 \end{aligned}$$

♦ Mean mark 37%.

13. Measurement, STD2 M1 2019 HSC 8 MC

$$\begin{aligned}
 \text{Absolute error} &= \frac{1}{2} \times \text{precision} \\
 &= \frac{1}{2} \times 0.1 \text{ kg} \\
 &= \frac{1}{2} \times 100 \text{ grams} \\
 &= 50 \text{ grams} \\
 \Rightarrow B
 \end{aligned}$$

♦ Mean mark 46%.

14. Measurement, STD2 M1 2020 HSC 2 MC

$$0.002073 = 2.073 \times 10^{-3}$$

$$= 2.1 \times 10^{-3} \text{ (to 2 sig fig)}$$

♦♦ Mean mark 35%.
COMMENT: A notable knowledge gap flagged by this result.

$\Rightarrow D$

15. Measurement, STD2 M1 2020 HSC 5 MC

$$\text{Absolute error} = 0.05$$

$$\% \text{ error} = \frac{0.05}{16.0} \times 100$$

$$= 0.3125 \%$$

♦ Mean mark 41%.

$\Rightarrow A$

16. Measurement, STD2 M1 2009 HSC 12 MC

$$\text{Since } 1 \text{ m}^2 = 100 \text{ cm} \times 100 \text{ cm}$$

$$= 10\,000 \text{ cm}^2$$

$$\therefore 0.0075 \text{ m}^2 = 0.0075 \times 10\,000$$

$$= 75 \text{ cm}^2$$

♦♦♦ Mean mark 19%.

$\Rightarrow C$

17. Measurement, STD1 M1 2020 HSC 3 MC

$$0.002073 = 2.073 \times 10^{-3}$$

$$= 2.1 \times 10^{-3} \text{ (to 2 sig fig)}$$

♦♦♦ Mean mark 12%.

$\Rightarrow B$

18. Measurement, STD2 M1 2012 HSC 26g

$$\text{Cups used per week} = 3 + 6 = 9$$

$$\text{Chlorine usage per week} = 9 \times 250\text{g}$$

$$= 2250\text{g}$$

$$\text{Total chlorine available} = 35\text{kg} = 35\,000 \text{ grams}$$

$$\text{Time it will last} = \frac{35\,000}{2250}$$

$$= 15.555\dots$$

MARKER'S COMMENT: Better answers converted all measurements to grams (rather than use decimals and kgs) and realised the answer should be in *full weeks*.

$\therefore$ The container will last 15 full weeks.

19. Measurement, STD2 M1 2008 HSC 23b

$$\text{Absolute error} = 5 \text{ mL}$$

$$\therefore \% \text{ error} = \frac{5}{1250} \times 100$$

$$= 0.4\%$$

20. Measurement, STD2 M1 2013 HSC 27d

i. Longer side = 25 cm

$$\text{Absolute error} = 0.5 \text{ cm}$$

$$\% \text{ Error} = \frac{0.5}{25} \times 100$$

$$= 2\%$$

♦♦ Mean mark 23%
MARKER'S COMMENT: Be aware that measurements accurate to the nearest cm have an **absolute error** for calculation purposes of 0.5 cm.

ii. Area = $l \times b$

$$\text{Area (upper)} = 25.5 \times 17.5$$

$$= 446.25 \text{ cm}^2$$

♦ Mean mark 35%

$$\text{Area (lower)} = 24.5 \times 16.5$$

$$= 404.25 \text{ cm}^2$$

$\therefore$ Area is between 404.25 cm^2 and 446.25 cm^2 .

21. Measurement, STD2 M1 2009 HSC 25b

We need to convert 50 mg into grams

$$50 \text{ mg} = \frac{50}{1000} = 0.05 \text{ g} = 5 \times 10^{-2} \text{ grams}$$

$$\begin{aligned} \therefore \text{Mass of 1 microbe} &= \frac{\text{mass of sample}}{\# \text{ microbes}} \\ &= \frac{5 \times 10^{-2}}{2.5 \times 10^6} \\ &= 2 \times 10^{-8} \text{ grams} \end{aligned}$$

♦♦♦ Mean mark 20%.

IMPORTANT: Can you solve: 8 apples weigh 1 kg, what does 1 apple weigh? This is **exactly the same concept**.